Policy Brief — Energy & Macroeconomic Analysis

Nuclear Energy Development and Inflation Dynamics in the United States

Dwanjai Oprien March 2026

Executive Summary

U.S. nuclear energy generation expanded rapidly from the 1970s through the early 2000s before stabilizing into a mature production system. Over the same period, core inflation shifted from volatility to long-term stability. The analysis finds no clear evidence that inflation directly influenced nuclear generation outcomes. Instead, inflation reflects the broader macroeconomic environment in which long-term infrastructure investment decisions occur.

Key Findings

- Nuclear generation increased substantially from 1970 to the early 2000s, followed by long-term stabilization.
- Core inflation was volatile during the 1970s and early 1980s, then stabilized in subsequent decades.
- No consistent relationship is observed between inflation trends and nuclear generation output.
- Nuclear energy development is driven primarily by structural and institutional factors rather than short-term macroeconomic fluctuations.

Analysis

Nuclear generation in the United States exhibits two distinct phases: a period of expansion followed by long-term stabilization. As shown in Figure 1, output increased steadily from the 1970s through the early 2000s, reflecting large-scale capacity development, before plateauing in subsequent decades.

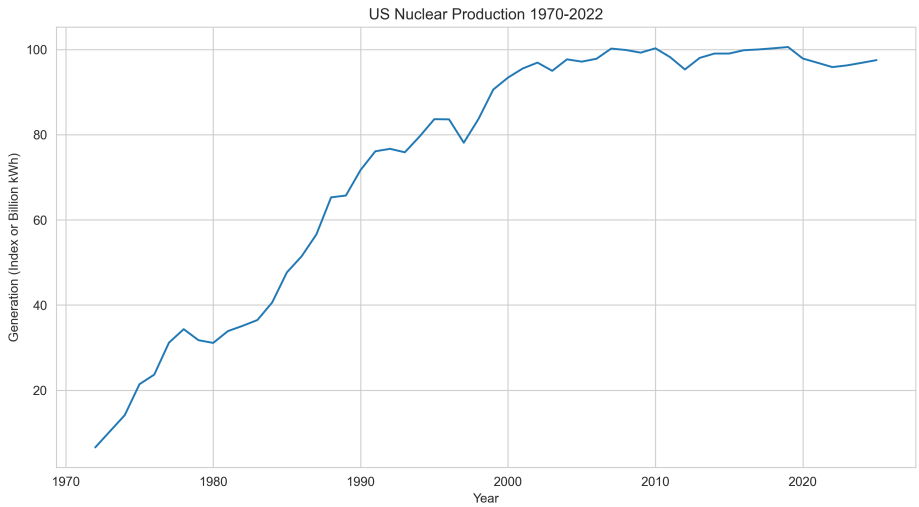

Figure 1. Annual average U.S. nuclear electric power generation from 1970 to 2022. The series shows a period of rapid expansion from the mid-1970s through the 1990s, followed by a prolonged plateau beginning in the early 2000s. This pattern reflects the transition from large-scale capacity buildout to a mature generation system with limited new plant additions and incremental efficiency gains

Figure 1: **Figure 1.** Annual average U.S. nuclear electric power generation, showing expansion followed by stabilization.

Core inflation, measured as the year-over-year change in Core CPI, was elevated and volatile during the 1970s and early 1980s before stabilizing at lower levels. As shown in Figure 2, inflation dynamics shifted significantly over time, reflecting broader macroeconomic transitions.

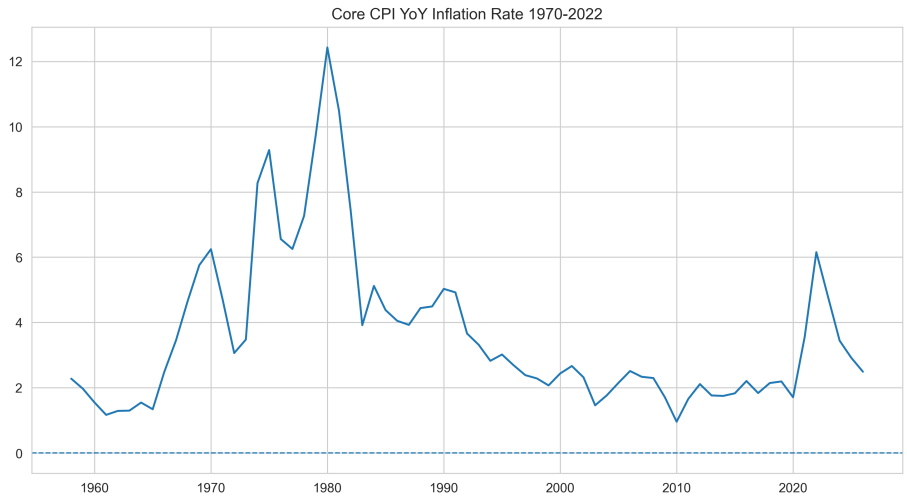

Figure 2. Annual average year-over-year Core CPI inflation from 1970 to 2022. Inflation was elevated and volatile during the 1970s and early 1980s, followed by a period of relative stability from the 1990s through the 2010s. A notable increase appears after 2020, reflecting renewed macroeconomic pressures. Core CPI is used to capture underlying inflation trends, excluding short-term volatility from food and energy prices.

Figure 2: **Figure 2.** Annual average Core CPI inflation (year-over-year), highlighting volatility in earlier decades and stability in later periods.

A comparison of these trends suggests that nuclear generation is not closely tied to short-term

inflation dynamics. Nuclear output continued to expand during periods of high inflation and remained stable during periods of lower inflation.

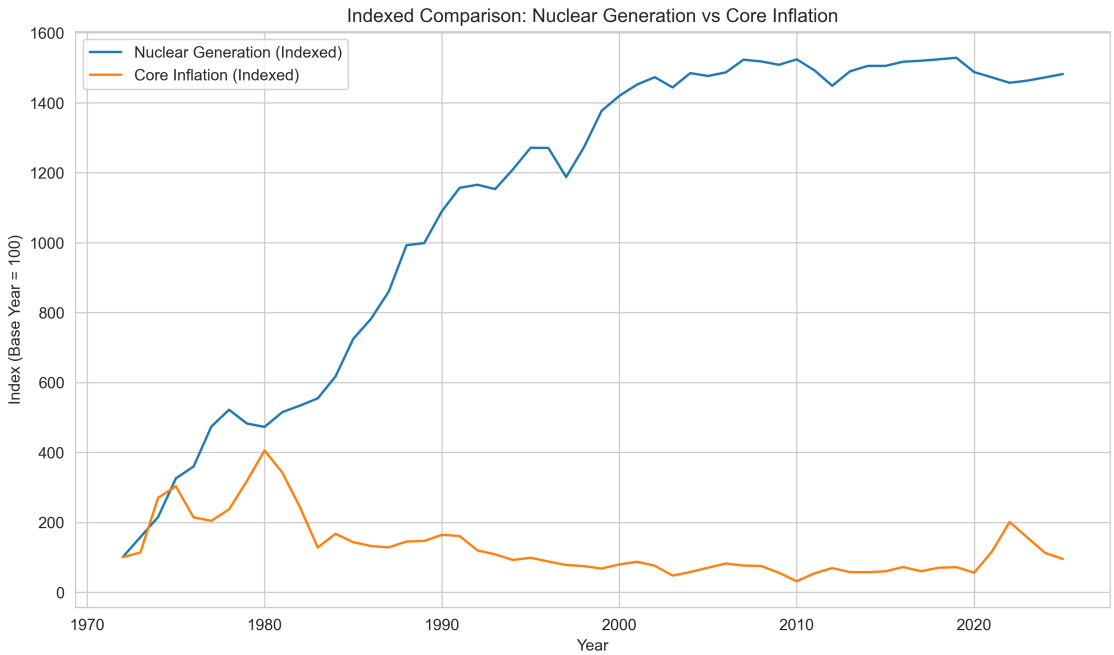

Figure 3: **Figure 3.** Indexed comparison of nuclear generation and core inflation (base year = 100), highlighting differences in long-term dynamics rather than absolute levels

These patterns reflect the structural characteristics of nuclear energy systems. Development is shaped by high upfront capital costs, regulatory constraints, and long planning horizons. As a result, nuclear generation is less sensitive to short-term macroeconomic fluctuations, including inflation.

Policy Implications

- Inflation should be interpreted as a contextual macroeconomic factor rather than a primary driver of nuclear generation outcomes.
- Long-term infrastructure systems such as nuclear energy are more strongly influenced by institutional, regulatory, and financing structures than by short-term price fluctuations.
- Policy analysis should prioritize structural and investment conditions when evaluating energy system development.

Data and Methods

This analysis uses data from the Federal Reserve Economic Data (FRED) database. Monthly nuclear generation and Core CPI data were converted into annual averages. Core CPI was used to calculate year-over-year inflation, which was then aggregated annually to align with nuclear generation data.